

Summary

I am a computational social scientist dedicated to understanding health inequalities with advanced AI algorithms.

My formal training includes degrees achieved at merit and distinction level in Sociology and Demography, Computational Econometrics, and Computer Science. Currently, I am a third-year DPhil student in Sociology at the University of Oxford. Alongside my doctoral research, I serve as a dedicated Research Assistant and consultant at the Leverhulme Centre for Demographic Science and the Department of Sociology at Oxford.

My research focuses on advanced machine learning, with particular emphasis on model evaluation and feature engineering.

All of my independent work involves applying machine learning algorithms, supported by a strong foundation in Mathematics, Statistics, and Econometrics. I am an advanced programmer with Python, R, Stata, and JavaScript expertise. I am proficient across the entire data science workflow—from data mining and cleaning to model development, parameter tuning, interpretability, and the deployment of interactive dashboards. I also collaborate on research software projects, including contributing to the development of RobustiPy, a Python package for multi-universal analysis.

Education

DPhil in Sociology	OXFORD, UK
University of Oxford	2022-Present
- Fully Funded by ESRC Grand Union Doctoral Training Partnership (DTP) Advanced Quantitative Methods (AQM) Awards - Leverhulme Centre for Demographic Science Graduate Scholarship, 2024/25	
MPhil in Sociology and Demography	OXFORD, UK
University of Oxford	2020-2022
- Passed with Merit - Core courses: Demographic Analysis, Life Course Research, Advanced Quantitative Methods, Research Design, Replication, Social Network Analysis.	
MSc Financial Economics	BIRMINGHAM, UK
University of Birmingham	2018-2019
- Passed with Distinction - Specialisation in Computational Econometrics - Nominated student representative - Core courses: Econometrics with Financial Applications, Macroeconomics, Theory of Finance - Dissertation title: "Combination forecasting based on American macroeconomic activity"	
BSc Computer Science and Technology	CHENGDU, CHINA
Southwestern University of Finance Economics	2014-2018
2015 -- 3rd Prize of Guanghua Entrepreneurship Competition of SWUFE 2016 -- Best Vice President of Arts Enterprise Management Co., Ltd. 2017 -- 2nd Prize of the Fifteenth Science, Technology and Culture Festival of SWUFE 2016-2018 -- Recipient of the Innovation and Entrepreneurship Award of Computer Science SWUFE (six times). - Core Courses: Core Programming (C), Object-Oriented Programming (JAVA), Object-Oriented Programming (JAVAEE), Data Structures, Computer Algorithms and Analysis, Principles and Applications of Databases, Large Data Processing, Data Mining and Applications, Mobile Technology Development	

Additional Training/ Affiliation/ Research Visits

The International Max Planck Research School for Population, Health and Data Science (IMPRS-PHDS)	
Max Planck Institute for Demographic Research	ROSTOCK, GERMANY
- Affiliated Member - Core Courses: Demographic Methods, Digital and Computational Demography, Population Health	
ESRC Center For Care	SHEFFIELD, UK
UKRI, NIHR and university collaboration	2022-present
Affiliated Member	

Office of Population Research
Princeton University

PRINCETON, US
2025

- Research Visit hosted by Matthew Saganik, Alexander Stewart 1886 Professor of Sociology.

Helsinki Institute for Demography and Population Health
University of Helsinki

HELSINKI, FINLAND
2024

- Research Visit hosted by Pekka Martikainen, Professor of Demography.

Summer Institute in Computational Social Science
University of Oxford

OXFORD, UK
2021

- Core Courses: Natural Language Processing, Machine Learning, Computer Vision, Web-scraping, Geospatial Analysis, Data Visualisation.
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Research Experience

University of Oxford

OXFORD, UK

Research Assistant at the Leverhulme Center for Demographic Science

GWAS Diversity Monitor

2021-2022

Project Leaders:

Melinda Mills, Nuffield Professor of Sociology, Director of LCDS, University of Oxford;

Charles Rahal, Associate Professor of Social Data Science, University of Oxford

- Dockerised Database construction.
- Maintenance of HTML and content.
- Debugging and enabling the function of Google Analytics on the website to track each component's visits and click-through rates.
- Debugging download function.
- Debugging and rewriting various figure functions to ensure it work with all possible drawing pathways.

A Scientometric Review of Population Studies

2021

Project Leaders:

Melinda Mills, Nuffield Professor of Sociology, Director of LCDS, University of Oxford;

Charles Rahal, Associate Professor of Social Data Science, University of Oxford

- Manually reviewed over 800 pieces of population-related literature and classified them into topics.

COVID-19 Vaccine Deployment Project

2020

Project Leaders:

Melinda Mills, Nuffield Professor of Sociology, Director of LCDS, University of Oxford;

Charles Rahal, Associate Professor of Social Data Science, University of Oxford

- Harmonising and filtering the literature on human vaccination in a corpus of over 20,000 documents.
- Contributing to the creation of a choropleth map of vaccination attitudes (effective, important and safe)

Research Assistant at the Nuffield College

Gun Violence Research

2021-2025

Project Leader: David Kirk, Professor of Criminology, University of Pennsylvania

- Built a scraper to recursively and repeatedly fetch information on guns for sale from armslist.com.
- Deployed the scraper to the Amazon Web Services and set up a dockerised PostgreSQL database.
- Maintenance of the scraper: building a monitoring system to supervise daily data collections.

Research Assistant at the Department of Sociology

Digital Gender Gaps

2021-present

Project Leader: Ridhi Kashyap, Professor of Computational Social Science, University of Oxford

- Harmonising Chinese prefecture-level development indicators (such as education and medical resources)
 - Facebook Marketing API Collector: Built a program to retrieve daily and monthly active audiences with different age groups and device information from Facebook Marketing API, worldwide.
 - Dockerised PostgreSQL database and Flask APP construction.
 - Project Pipeline construction, from downloading, preprocessing and analysing the daily collected data to uploading the results to a database linked to the back-hand of a web application.
 - Database daily maintenance.
 - Sub-national data collection and customised tessellation for Low- and Middle-Income Countries.
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Teaching Experience

Department of Politics and International Relations, University of Oxford

OXFORD, UK

Co-lecturer: Intro to Statistics

2023-2024

- Facilitated two R laboratory workshops throughout the course for postgraduate students, focusing on introducing R as an essential tool for statistical analysis.

Publications and Working Papers

Published Works

- Leasure, D.R., Kashyap, R., Rampazzo, F., Dooley, C.A., Elbers, B., Bondarenko, M., Verhagen, M., Frey, A., **Yan, J.**, Akimova, E.T., Fatehkia, M., Trigwell, R., Tatem, A.J., Weber, I. and Mills, M.C. (2023). Nowcasting Daily Population Displacement in Ukraine through Social Media Advertising Data. *Population and Development Review*. <https://doi.org/10.1111/padr.12558>
- **Yan, J.**, Rahal, C. On the unknowable limits to prediction. *Nature Computational Science* 5, 188–190 (2025). <https://doi.org/10.1038/s43588-025-00776-y>

Policy Reports

- Mills, M., Rahal, C., Brazel, D., **Yan, J.**, and Gieysztor, S. (2020). COVID-19 Vaccine Deployment: Behaviour, ethics, misinformation and policy strategies. *London: The Royal Society & The British Academy*.

Software and Tools Development

- Leasure DR, **Yan, J.**, Bondarenko M, Kerr D, Fatehkia M, Weber I, Kashyap R. (2023). Digital Gender Gaps Web Application, v1.0.0. Zenodo, GitHub. doi:10.5281/zenodo.7897491. <https://github.com/OxfordDemSci/dgg-www>.
- Valdenegro, D., **Yan, J.**, Dai, D., Rahal, C. (2025). *RobustiPy: A Python Package for Robustness Checks in Social Science Research*, v0.1.0. Zenodo, GitHub. doi:10.5281/zenodo.15700698. <https://github.com/RobustiPy/robustipy>

Under Review

- Breen, C. F., Fatehkia, M. **Yan, J.**, Zhao, X. Leasure, D. R., Weber, I., Kashyap, R. (2024) ‘Mapping subnational gender gaps in internet and mobile adoption using social media data’, peer-reviewed and revised at *Proceedings of the National Academy of Sciences of the United States of America*.
- **Yan, J** (2025). ‘Revisiting the social determinants of health with explainable AI: a cross-country perspective’, peer-reviewed and revised at *American Journal of Epidemiology*.
- Valdenegro, D., **Yan, J.**, Dai, D., Rahal, C. (2025), ‘Introducing RobustiPy: An efficient next generation multiversal library with model selection, averaging, resampling, under reviewing at *Nature Computational Science*. <https://arxiv.org/abs/2506.19958>

Working Papers

- Rahal, C., **Yan, J.**, M., Verhagen, (2025). ‘From a Single Seed Grows a Forest of Uncertainty’, in progress/preparation for submission to leading journals and conferences.
- **Yan, J** (2025). ‘Bridging Social Factors to Pathological Outcomes: A Deep-Learning Pseudo Imaging based Approach’, in progress/preparation for submission to leading journals and conferences.
- **Yan, J.**, Li, J., Breen, C.F., Kashyap R. (2025). ‘Nowcasting global digital gender gaps using social media data’, in progress/preparation for submission to leading journals and conferences.

Conferences

- 9th International Conference on Computational Social Science (IC2S2) COPENHAGEN, DENMARK
Poster Session, presenting work: Predicting death Holistically and Precisely with Explainable AI
- PAA 2024 Annual Meeting COLUMBUS, OHIO
Poster Session, presenting work: Predicting death Holistically and Precisely with Explainable AI
- European Population Conference 2024 EDINBURGH SCOTLAND, UK
Oral presentation in session Machine Learning Approaches for Population Research, presenting work: Death Predictions from Social Determinants: Holistically and Precisely with Explainable AI
- Demystifying Machine Learning workshop 2024 by MPIDR ROSTOCK, GERMANY
Oral presentation in session Machine Learning Approaches for Population Research, presenting work: Death Predictions from Social Determinants: Holistically and Precisely with Explainable AI
- PAA 2025 Annual Meeting WASHINGTON D.C., US
Poster Session, presenting works: Discovering Prevalent Chronic Disease Profiles with Advanced Clustering Methods; Nowcasting Global Digital Gender Gaps Using Social Media Data

Languages and Skills:

• Python	• Java	• IELTS	• Eviews	• Japanese (N3)
• R	• JavaScript	• Stata	• HTML	• L ^A T _E X

Extra-curricular Activities

European Association of Population Study	Netherlands
Steering Committee Member	<i>2024-present</i>
• Organising events for and with PhD students and early career researchers in Europe and in collaboration with EAPS.	
• Organising early career/student events and webinars.	

Professional History

Campus President, Chengdu Arts Enterprise Management Co., Ltd.	Chengdu, China
Campus President	<i>2016</i>
• Supervised daily operations.	
• Took charge of the management of SWUFE intra-campus company operation.	
• Acted as the main coordinator of musical instrument training and operation.	
• Attracted and encouraged engagement and participation.	
